



Reading: Introduction to AI for IT Support

In the fast-paced world of IT support, providing timely and effective assistance requires leveraging the latest technology. AI tools are transforming the way IT support professionals work, offering powerful capabilities to streamline ticket management, automate communications, and proactively monitor systems.

Artificial intelligence (AI) refers to computer programs that can complete cognitive tasks typically associated with human intelligence. You can use AI tools to augment and automate general IT support tasks, such as drafting responses to common inquiries, summarizing technical information, and helping to triage support tickets.

In this lesson, you'll discover how you can integrate AI into your daily IT support activities. We'll explore how AI is already being used by IT support professionals to help automate routine tasks, enhance productivity, and improve the overall user experience.

Throughout this lesson, you will:

- Learn foundational concepts of AI.
- Discover AI tools used in the IT support field.
- Review examples of how AI is used in the day-to-day work of a Google IT support professional.
- Gain hands-on practice in using AI to streamline your tasks.

As you begin your career as an IT professional, this lesson will give you the basic information you need to experiment with AI tools, identify opportunities for leveraging AI in your IT support role, and keep up to date with industry changes.

Support and maintain AI systems

As AI systems become more commonly used in professional fields, it will be part of your job as an IT professional to maintain and protect those systems for your organization. AI systems are complex and rely on large quantities of high-quality data to operate well. For that reason, it's helpful to understand what qualities make AI systems unique and challenges they may present for you as an IT professional. AI systems are unique because they are:

- **Intricate:** Many AI systems are complex, making it hard to understand how they work. For this reason, troubleshooting issues and diagnosing errors may be more difficult with AI systems. However, developing expertise about the architecture of AI can set you apart as an IT professional who knows how to address difficult issues with IT systems.
- **Iterative:** Unlike traditional software, AI systems constantly iterate, or change, so the behavior of the system may vary over time, making it difficult to maintain documentation and predict

future issues. This is why it's important for you to stay up-to-date with the latest AI technologies, as well as updates to existing AI technologies.

- **Data dependent:** The performance of an AI system depends on the quality and quantity of its training data. As an IT professional, you will maintain this data. For example, you will likely need to determine where the data is stored and how it's secured.
- **Cybersecurity targets:** Because AI systems rely on a lot of high-quality data to operate, they can become the targets of cybersecurity attacks from malicious actors. AI systems might require additional security measures and expertise to protect them. As an IT professional, you should understand potential attack vectors, as well as how to implement robust defenses, so that you can integrate these technologies safely within your organization's existing systems.
- **Complicated to integrate:** Making sure AI systems work with existing IT infrastructure can be difficult, requiring compatibility checks and adjustments to workflows. This can be particularly challenging when you need to integrate AI systems with legacy systems. But by developing your expertise with AI technologies, you can help your organization integrate AI systems and stay at the forefront of innovation.

Because many of these challenges are unique to AI systems, you will need to develop additional skills as an IT professional. Fortunately, the AI tools themselves can help you do that. AI tools can help you research information about AI, troubleshoot unique errors, and create documentation to reference in the future. As you continue to learn about how AI can be used in IT automation, consider how you can leverage AI tools to help you overcome challenges you might encounter as an IT professional.
